

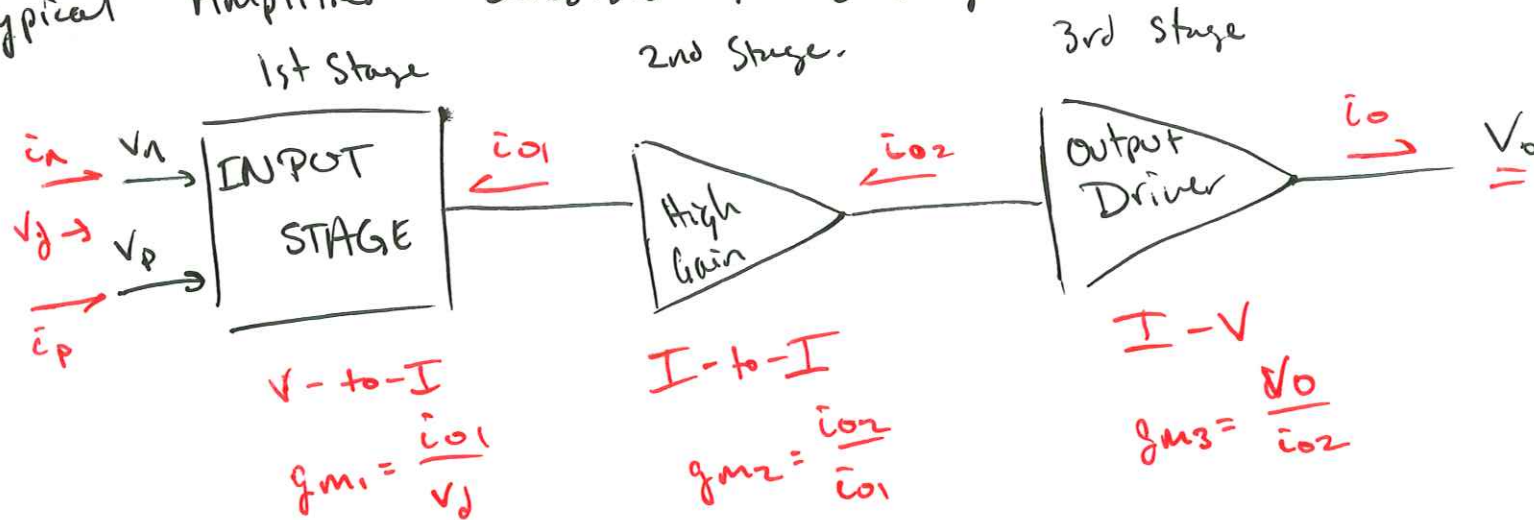
1st Exam Next Thursday 2/19

10:00 - 11:50

Open-Book, Open Notes (computer open, No typing)

HW 1-4, Lectures 1-7, Book Ch. 1-2

Typical ^{OP}-Amplifier consists of 3 stages:



typically
 $g_{m2} \approx 10^4 - 10^7$

$$g_{m1} g_{m3} = \frac{v}{V} \approx 1$$

$I_A, I_C \Rightarrow$ constant current sources

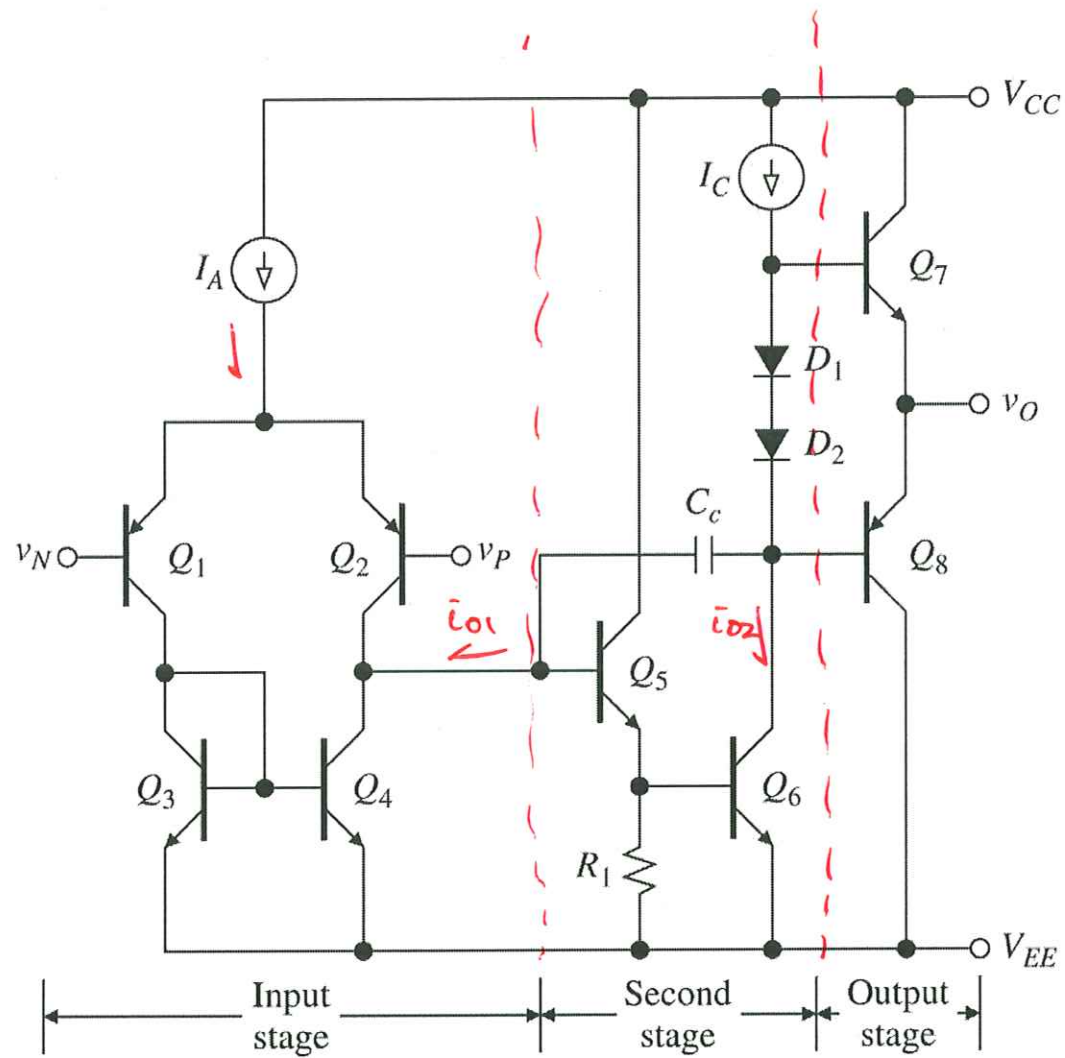


Figure 5.1: Simplified circuit diagram of a typical bipolar op amp.

$$\omega_1 = I_A \tanh\left(\frac{v_D - v_n}{v_T}\right)$$

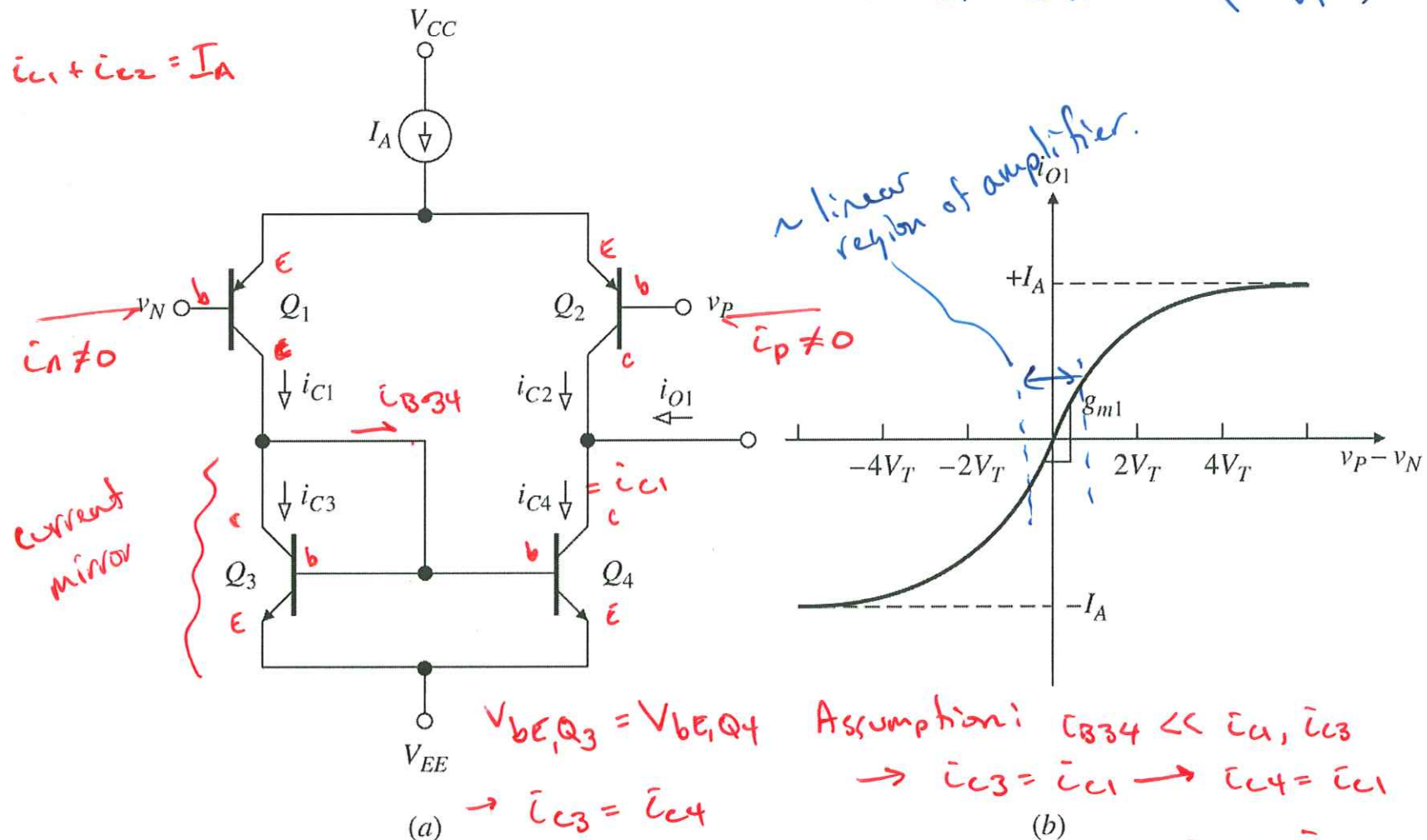


Figure 5.2: Input stage and its transfer characteristic.

$$\bar{c}_4 = \bar{c}_2 + \bar{c}_1 = \bar{c}_1$$

$$\hookrightarrow \bar{c}_1 = \bar{c}_1 - \bar{c}_2$$

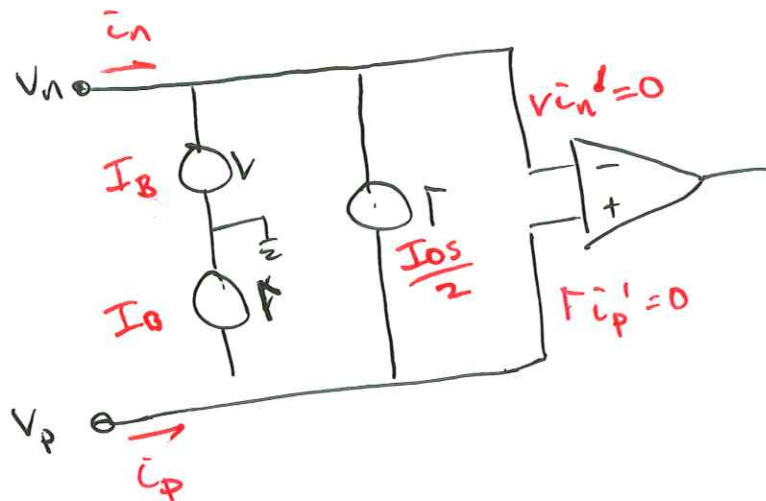
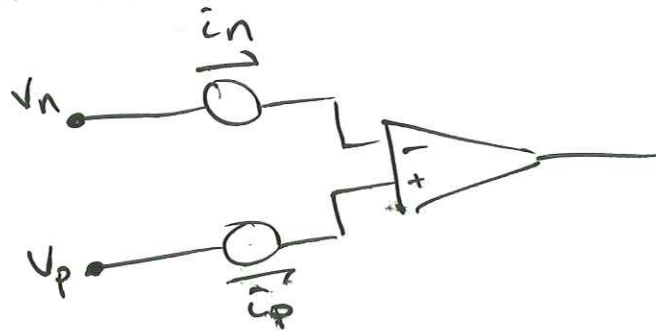
$\uparrow$
prop. to v_n

$\uparrow$
prop. to v_p

Non-Zero Input Currents:

$$\bar{i}_n \neq 0, \bar{i}_p \neq 0$$

Model as external sources:



$$\bar{i}_n \approx \bar{i}_p \approx I_B$$

I_B is transistor Bias current.

$$\bar{i}_n \neq \bar{i}_p \text{ typically } \neq$$

why: manufacturing defects
thermal gradients

$$\bar{i}_n = I_B - I_{OS}/2$$

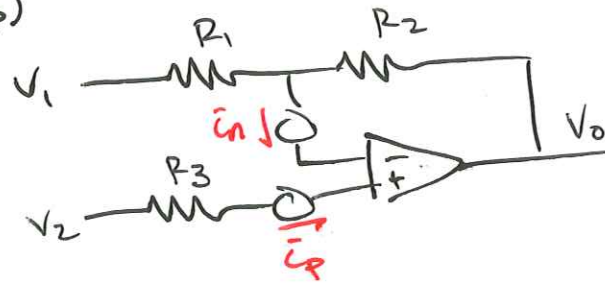
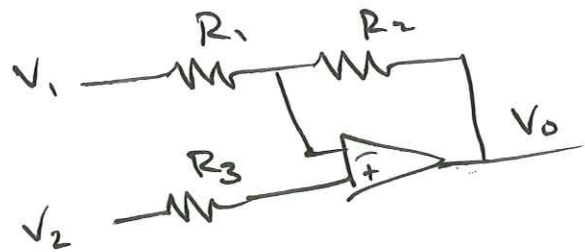
$$\bar{i}_p = I_B + I_{OS}/2$$

$I_{OS} =$ ~~offset~~ offset current

→ typically stated as a $\pm$ value

$$I_B = \frac{\bar{i}_p + \bar{i}_n}{2} \quad I_{OS} = \bar{i}_p - \bar{i}_n$$

Error Induced by I_B & I_{OS} (i_n, i_p)



4 independent sources (V_1, V_2, i_n, i_p)

Superposition:

$\alpha \rightarrow \infty, r_o \rightarrow 0, r_d \rightarrow \infty$

① $V_1 \neq 0, V_2, i_n, i_p = 0$

$\rightarrow$ inverting amplifier

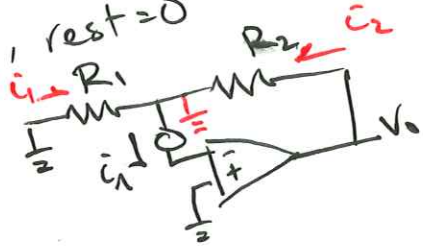
$$V_{O1} = -\frac{R_2}{R_1} V_1$$

② $V_2 \neq 0, \text{rest} = 0$

$\rightarrow$ non-inv. amp.

$$V_{O2} = \left(1 + \frac{R_2}{R_1}\right) V_2$$

③ $i_n \neq 0, \text{rest} = 0$

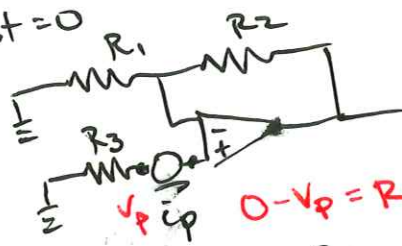


$$i_1 = 0$$

$$i_2 = i_n$$

$$i_2 = i_n = \frac{V_{O3} - V_n}{R_2} \Rightarrow V_{O3} = R_2 i_n$$

④ $i_p \neq 0, \text{rest} = 0$



Non-inv. amp
with $V_{in} = V_p = -R_3 i_p$

$$V_{O4} = -R_3 \left(1 + \frac{R_2}{R_1}\right) i_p$$

$$V_O = \underbrace{-\frac{R_2}{R_1} V_1 + \left(1 + \frac{R_2}{R_1}\right) V_2}_{\text{ideal operation}} + \underbrace{R_2 i_n - R_3 \left(1 + \frac{R_2}{R_1}\right) i_p}_{\text{non-ideal operation.}}$$

from previous: $V_O = V_{O_{Ideal}} + V_{OI}$
↑ output due to input currents

$$V_{OI} = R_2 i_n - R_3 \left(1 + \frac{R_2}{R_1}\right) i_p$$

$$V_{OI} = \underbrace{\left(R_2 - R_3 \left(1 + \frac{R_2}{R_1}\right)\right)}_{\text{can drive this to 0!}} I_B - \underbrace{\left(R_2 + R_3 \left(1 + \frac{R_2}{R_1}\right)\right)}_{\text{makes } I_{os} \text{ contribution larger!}} I_{os}$$

↑
can drive this to 0!

$$\rightarrow R_3 = \frac{R_1 R_2}{R_1 + R_2} = R_1 \parallel R_2$$

Drawback: makes I_{os} contribution larger!

$R_3 = 0$, smallest I_{os} contribution.

Typically, $I_B > I_{os}$, more beneficial to use $R_3 = R_1 \parallel R_2$

Ex. 741, $I_B = 80 \text{ nA}$, $I_{os} = \pm 20 \text{ nA}$

let $R_1 = 10 \text{ k}$, $R_2 = 50 \text{ k}$, $R_3 = 0$ ~~$\rightarrow R_3 = 0$~~

$$V_2 = 0$$

$$\rightarrow A = -5 \text{ V/V}$$

$$V_{OI} = 4 \text{ mV} \pm 1 \text{ mV}$$

→ Range: $3 \text{ mV} \leftrightarrow 5 \text{ mV}$

Change $R_3 = R_1 \parallel R_2 = 8.3 \text{ k}\Omega$

$$V_{OI} = \pm 2 \text{ mV}$$

→ Range: $-2 \text{ mV} \leftrightarrow 2 \text{ mV}$

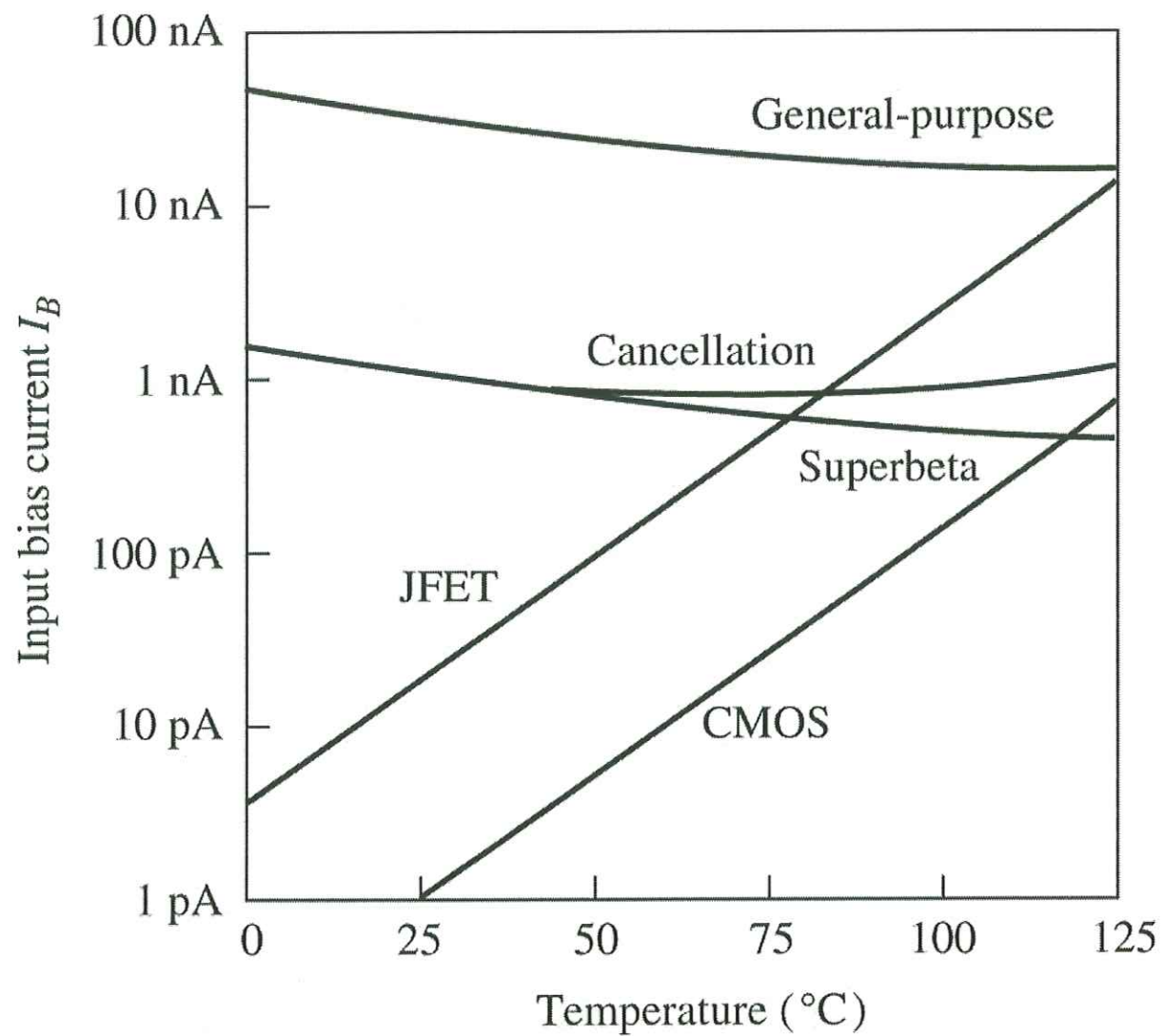


Figure 5.10: Typical input bias current characteristics.